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Connectivity evaluations of Indian Railways zones with superfast express (SE) trains

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Abstract

A high percentage of rail ridership in India is facilitated by the superfast express (SE) trains that operate across and within the 17 zones of the Indian Railways. While large cities with SE train stations in zones have some form of first/last-mile transport connectivity and considered safe to access during nighttime or early wee hours, several other stations do not have this privilege. Therefore, considering the 9 pm – 7 am period as not so friendly hours of travel, and the 7 am – 9 pm period as the desired hours of travel, analysis shows that the inequalities in centrality and accessibility exist between the zones more than between the stations within a zone for these two periods. Further, it was found that the two zones of the Northern Railway and the Southern Railway have the largest percentage of stations with deterioration in centrality for the 7 am – 9 pm period compared to the 24-hour period. If evaluated for accessibility, there were five zones of the South Western Railway, Southern Railway, East Coast Railway, South Eastern Railway, and Northern Railway that would experience deterioration in accessibility for the same desired hours of travel. It was also found that the investment made by the Indian Railways across the 17 zones between 2016 - 2020 resulted in positively impacting the total employment growth in the zones, and it was significant. The mean accessibility of stations in a zone, served by SE trains, had a significant positive impact on the zonal job opportunities during the 7 am – 9 pm period. The mean accessibility for the 9 pm – 7 am period showed a significant negative impact on job opportunities. These findings suggest that the Indian Railways should continue investing and improving zonal accessibility with new SE trains, particularly during the 7 am – 9 pm period, to spur economic growth in the zones. In addition, such investments with new SE trains should be appropriately planned to reduce disparity among zones.

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1. Introduction and Background

Passenger rail connects spatially far apart places and plays a vital role in attenuating regional disparities (Fraszczyk et al., 2016). Rail as a transit system can address social equity issues and justice by allowing transport access to those who are poor or in more disadvantaged positions (Lu, 2018). Rail has been known to reduce per capita travel costs and

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boost economic growth both in developed nations such as the United States of America (Litman, 2015) and developing nations such as India (Pradhan and Bagchi, 2013). Numerous studies underline the economic development benefits of rail transportation. Economic development can assume various forms, including business growth, employment, and income increase. Rail attempts to achieve these forms due to the connectivity and access it facilitates between two large cities (Litman, 2015). However, this is not always the case. Past studies on economic agglomeration on a spatial scale indicate that an increase in regional disparities can result from an increase in infrastructure investments and connectivity – this occurs when the more prominent cities can now draw on increased resources from the new connected smaller cities (Fujita and Thisse, 1996). Thus, connectivity between rich and poor regions may not always work out to be as mutually beneficial (Liu et al., 2020).

The superfast express (SE) trains in India are a unique candidate for study. They connect the country's vast socio-economic landscapes and operate day and night (Ravi, 2019). Some SE trains are always crowded and marred with delays but are quite popular in India (Kumar et al. 2018; India.com, 2020). However, very little is known on the quality of connectivity provided by the SE trains when compared for day versus night service (i.e., the odd travel hours). This is important as policy decisions on investments in rail network expansions/improvements can occur accordingly to minimize regional disparities through SE train connectivity. Recent studies have focused on understanding the regional disparity (or integration) and economic impacts of the rail to a great extent – however, in the context of high-speed rails (Liu et al., 2020; Chandra and Vadali, 2014). The same for the SE trains has seldom been carried out for the role they may play (or maybe not?) in improving the socio-economic disparities in developing nations like India.

This paper evaluates the service provided by the SE trains across the 17 zones of the Indian Railways using a new formula for station connectivity evaluations for the two periods - the standard 24-hour period and the odd travel hour period (involving nighttime to early wee hours). The analysis is carried out to understand a zone's contribution to job opportunities from the Indian Railways investment from 2016 to 2020.

The rest of the paper is structured as follows: Section 2 provides a review of pertinent literature. Section 3 presents the methodology adopted for SE train service network connectivity and procedure for evaluating the spillover effects of investment and accessibility for zones. Section 4 presents the analysis and findings on connectivity and zonal disparities. Section 5 discusses investment policy and other connectivity implications in zones, and section 5 concludes the research.

2. Literature Review

Respective rail authorities of various nations have undertaken improvements through investments in passenger rail with the primary goal of providing alternatives to highway travel and improving mobility (Wenner et al., 2020; Murakami and Cervero; Sitharamaraju et al., 2020). However, these improvements should always be in-sync with passenger expectations and perceptions of the rail system's quality of service (such as reduced delays and comfort) (Johnson, 2020; Choudhary et al., 2021). Eliminating delays by increasing rail speeds is a common step to improving service that can have varied impacts on intra- and inter-agglomeration mobility (Romashina, 2021). One of the motivations for improving rail service is to attract riders to the transit and sustain economic growth for a region (Chandra and Mazin, 2020).

Recently various surrogates of economic growth “potential” have been studied in the literature for rail transportation – but those that are scholarly have been limited mainly to high-speed rail. For example, the growth potential used in the context of internal accessibility, also known as ‘self-potential’, is significant for regions or zones with high employment and population served by the rail – again for high-speed trains (Chandra and Vadali, 2014). Despite the best efforts, we could not find a single study that is comprehensive enough to discuss the economic growth potential from the semi-high speed and express trains with lower speeds than the high-speed rail.

Similarly, on the other extreme, literature exists that assesses passenger rails' economic potential, such as light rail with a lower speed than semi-high speed and express trains. In one example study, Fan et al. (2012) found that accessibility increased among low-wage workers compared to medium and high-wage workers after implementing the light rail line service. Furthermore, the accessibility gains were spatially predominant in areas that exhibited the rail connected with bus routes, indicating that last-mile connectivity with bus service plays a vital role in rail success.

2.1. Connectivity Measures in Practice

Graph theory has been widely used in devising measures for transit connectivity. For example, Kaplan et al. (2014) used transit connectivity that considered various measures, including in-vehicle time, access/egress times, waiting time, service reliability, frequency, and ‘seamless’ transfers along multi-modal paths. Other studies have considered a host of connectivity measures such as node connectivity and line connectivity (Mishra et al., 2012). However, none of these and other studies that exist are targeted to assess service quality with specific time of the day considerations of the service accounting for first/last mile connectivity convenience in rail transportation.

The importance of rail connectivity in facilitating a satisfactory level of performance and service has been well acknowledged. For example, a study by Zhu et al. (2018) highlighted the significant contribution of train service in connectivity for 19 of the 20 top-ranking domestic routes in China, and Xu et al. (2019) noted rail connectivity to stations as critical to performance assessment of rail transportation.

One of the popular measures of network connectivity is centrality, generally used in the field of network theory, e.g., in physical networks, computer science, epidemiology etc. (Martínez et al., 2003; Liu et al., 2005; Bell et al., 1999; Junker et al., 2006; Guimerà et al., 2005). Although other network connectivity measures are also used in public transportation, centrality has only recently been proposed (Mishra et al., 2012). To measure centrality – often called the topological feature of a network – the “accessibility” measure has also been used (Wangtu et al., 2018; Liu et al., 2020). The premise is that focusing only on the topology of a network isn’t sufficient and passenger flow has to be considered in describing the appropriate connectivity measure for a network (Sun et al., 2018). In the context of rail, most studies we came across analyzed the impact of high-speed rail in identifying the winner or loser cities with accessibility (Moyano et al., 2018). However, minimal understanding exists on the centrality or accessibility impacts of SE trains on the cities and regions they serve.

Accessibility in public transport has been studied in forms, such as location and potential (Geurs and van Wee, 2004). The location-based accessibility is the simplest form expressed as travel time, distance, or cost requirements to access a destination (Karou and Hull, 2014). The potential-based accessibility measure accounts for the transportation network’s ability to facilitate access to a population or workforce (Saghapour et al., 2016; Chandra and Vadali, 2013). Most research adopts either of these two forms of accessibility for rail.

2.2. Social Equity and Disparity with Rail

Transit accessibility as a connectivity measure is usually used as a yardstick to measure transport equity (Ermagun and Tilahun, 2020). Spatial impact analysis can be used to evaluate equity implications derived from changes to the distribution of accessibility values in the specific urban agglomerations of the rail projects (Monzón, 2013). Proximity to population centers, the existence of other rail stations, and the quality of the overall transportation system impact these implications. With the service offered to high-density populations, rail services with high speeds are usually recommended for cities that serve as regional core (Martin, 1997). In contrast, towns outside such high-speed rail service suffer spatial disparities but benefit from SE trains by providing transport accessibility within lower-density areas (Kim and Sultana, 2015). Studies show that population groups have been at the helm of equity-related studies resulting from accessibility, i.e., accessibility has been studied in terms of its distribution among population groups by assessing the spatial equity of transport infrastructure projects (Bröcker et al., 2010).

Transportation equity is often variedly related and labeled by names such as justice and fairness (Bröcker et al., 2004; Monzón et al., 2013). An equitable transit system is often considered attractive to a more extensive society and for a sustained level of ridership. One of the efforts in research has been to measure public transport equity from a graph-theoretical approach (Welch and Mishra, 2013). In this paper, we adopt a similar approach to identify inequalities in connectivity across zones demarcated by the rail networks of the Indian Railways.

Policies that drive infrastructure investments in a city should reduce inequalities, which motivates disparity analysis (Liu et al., 2020). Connectivity as centrality and accessibility provided by rail transportation brings about economic changes, and it often requires policy interventions to distribute the economic developments across cities connected via stations appropriately. Three commonly used measures can be deployed for disparity evaluation, and these include dispersion indices such as coefficient of variation (Gutierrez, 2001), Gini coefficient based on Lorenz curve (Kim, 2000), and entropy index such as the Thiel’s index (Chen and Haynes, 2017).

The SE trains of the Indian Railways connect cities of varied population sizes and thus, facilitate the distribution of economic activities. This also results in uneven gains in growth for cities, thereby creating inequalities or disparities among them. Thus, an assessment of disparities helps determine appropriate policies needed to support cities to prosper economically and not be left behind from taking advantage of the benefits resulting from rail connectivity.

2.3. Rail Transportation in India

India's first documented rail operation began in 1853 (IR History, 2020), and since then, the railway network has expanded throughout the country, with 67,415 kilometers of route lengths in 2019. During 2018-19, the Indian Railways had recorded almost 8.5 billion passengers originating (i.e., passengers with their trip beginning with the train). The passenger-kilometers for suburban and sub-urban regions have been around 1.2 trillion (Ministry of Railways, 2020). The upper-class segment constituted about 179 million passengers originating and 1.26 billion passenger-kilometers. The cars in this class are primarily two-tiered sleeping or seating chairs found in most long-distance SE trains. There are currently around 1,200 such trains operating in India, including the Rajdhani Express, Durgam Express, Shatabdi Express, and other superfast trains of the Indian Railways (etrain.info, 2020). The average speeds of these trains are 85 - 95 km/hr (Indian Railways, 2020). Twenty-four pairs of Rajdhani Express trains operate in India and connect New Delhi, the capital of India, to other state capital cities of the country (Indianrailinfo, 2020). About 25 pairs of operational Durgam Express trains are known for their non-stop services between India's major cities and state capitals (etrain.info, 2020). The Durgam Express trains and the Rajdhani Express trains are similar in terms of fare and priority in operations. The Shatabdi Express trains are also superfast trains, and there are currently 25 of these operating between major cities – out of those, 15 trains originate from New Delhi (goibibo.com, 2020). These trains are mostly day-trains, i.e., they complete their journey on the same day after leaving their originating stations. Superfast trains have a higher number of halts, and the fare is lower than the three express trains of Shatabdi, Rajdhani, and Durgam. As per a report from the Indian Railway Year Book 2018-19, although the upper-class segment had a tiny percentage of share for passenger-kilometers, the revenue in earnings per passenger-kilometer was the highest from the upper-class (Indian Railways, 2020). There is an estimated investment of US\$ 715 billion through various projects in Indian railways by 2030 (IBEF, 2021), as well as investment from privately operated train services such as the Delhi-Lucknow Tejas Express (PTI, 2019).

In this paper, we examine the regional disparity in connectivity of the zones that divide the Indian rail service network of the Indian Railways. The seventeen Indian Railway zones are studied for connectivity using degree centrality and accessibility - both employed as indicators of economic potential. The zones studied are namely, the Central Railway, East Central Railway, East Coast Railway, Eastern Railway, North Central Railway, North Eastern Railway, North Western Railway, Northeast Frontier Railway, Northern Railway, South Central Railway, South East Central Railway, South Eastern Railway, South Western Railway, Southern Railway, West Central Railway, Western Railway, and the Konkan Railway. These zones are assumed to be economically significant regions of activity for the Indian Railways resulting from 1.2 trillion passenger-kilometers across SE train stations in India.

3. Methodology

A generic form of connectivity using degree centrality and accessibility - both employed as indicators of economic potential – have been proposed in this paper. However, a basic understanding of the rail transportation network has been developed first and discussed below.

3.1. Network Topology

The transportation network for rail can be divided into two parts – the physical network consisting of the rail tracks and the service network that comprises trains that service passengers across the network. While the rail tracks represent a fixed form of connectivity offered across stations, the service network indicates where passenger flows occur. In this context, a general form of rail infrastructure and service network can be defined based on the work by Li and Rong (2020). The rail network can be represented as a graph, $G = (V, T, S)$ where V is the stations of SE trains, T is the links or rail lines, with travel time information, used by SE trains. Subsequent details of the networks are discussed below.

3.1.1. Infrastructure Network

The infrastructure network is formed by enumerating all existing links of graph G with V stations and T links. In other words, the infrastructure network is formed by links between any two stations of the rail network connected by tracks. Fig. 1 shows the infrastructure network created by six stations A to F. The infrastructure network is independent of the number of trains serving these stations.

3.1.2. Travel Time Network

The travel time network represents travel time as the weight for each link between stations served by a train (see Fig. 1). It is assumed that this network has equal travel time in each direction for a link. The general representation of the travel time of a link is T_{ij}^k where k stands for the train number operating between stations i and j .

3.1.3. Service Network

All the links among stations V that are served by train(s) form the service network. The S links of the SE train service network are divided into parts S_I and S_{II} - representing Part-I and Part-II service networks (see Fig. 1). The Part-I service network is formed when the departure time of train k at the origin station i or the train k 's arrival time at destination j is within the desired travel periods. On the other hand, in Part-II, the arrival or the departure time of the train k at the origin station i (or the destination station j) is during the odd hours of travel that may not be convenient to passengers. Thus, travel in the Part-II service network is *not* considered friendly from a passenger point of view. For example, under Part-II, a nighttime or early morning hour travel (as odd travel times) might not be perceived safe by a passenger due to dark or a lack of first/last mile connectivity to stations.

Therefore, based on the above narrative, for a dummy variable, say $t_{i,j}^k = 1$ as in Part-I, it would indicate train k that serves the link connecting station i to j has an arrival or departure time at i (or at j) that is during the friendly travel periods. By default, $t_{i,j}^k = 0$ in Part-II service network.

From the network throughput perspective, entities managing the above-discussed networks aim to utilize as many links from the infrastructure network with as many trains as possible to create more extensive service networks. Policies that allow such efforts are generally encouraged, although operational budget constraints limit those efforts.

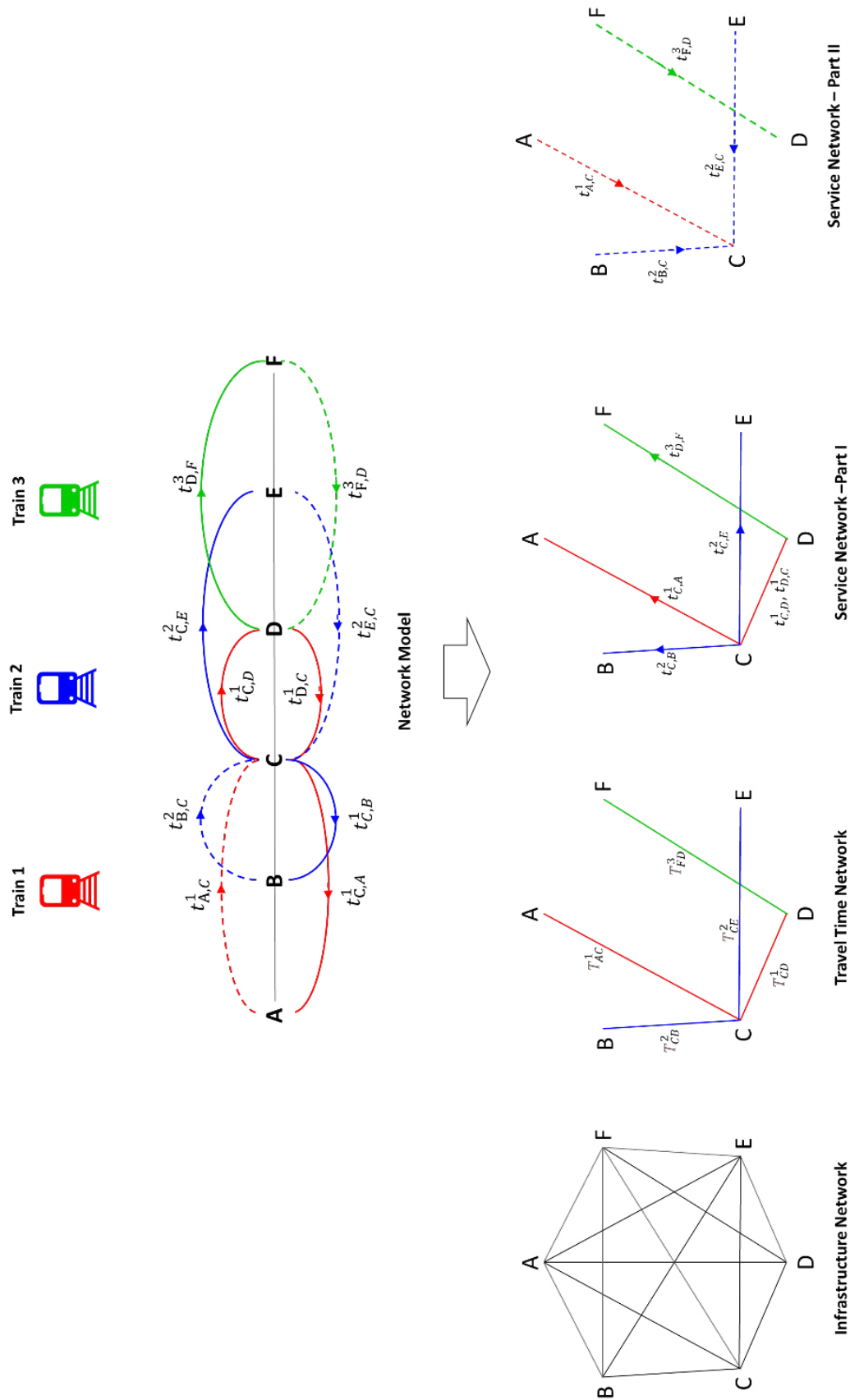


Figure 1: Network representations of the three-train service example

3.2. Connectivity

A formulation for connectivity is proposed for the two service networks discussed above. We use centrality and accessibility as the primary measures of connectivity of the service network.

3.2.1. Centrality

A modified form of “degree centrality” measure has been used in this research. The measure serves as the indicator of connectivity for stations and extends for zones that contain these stations. During late night to early morning wee travel hours, travel by rail becomes unfriendly especially due to lack of an adequate first/last mile transport connectivity or safety concerns of rail passengers at dark. Therefore, an appropriate degree centrality formula (C_i) of a station i is required to be developed for Part II railway service network shown in Fig. 1 and is expressed as:

$$C_i = \sum_{i \neq j \in N} t_{i,j,a}^k w_{ij}^k + \sum_{i \neq j \in N} t_{j,i,d}^k w_{ji}^k \quad (1)$$

where,

$$t_{i,j,a}^k = \begin{cases} 1, & \text{if a train } k \text{ departs station } i \text{ during desirable / friendly periods to reach station } j \\ 0, & \text{otherwise} \end{cases}$$

$$t_{j,i,d}^k = \begin{cases} 1, & \text{if a train } k \text{ departs station } j \text{ to arrive at station } i \text{ during desirable / friendly periods} \\ 0, & \text{otherwise} \end{cases}$$

$$w_{ij}^k = 1, \text{ if there exists a train } k \text{ from station } i \text{ to station } j, \text{ otherwise } 0.$$

$$w_{ji}^k = 1, \text{ if there exists a train } k \text{ from station } j \text{ to station } i, \text{ otherwise } 0.$$

3.2.2. Accessibility

Accessibility is often used as an indicator for determining economic potential (Chandra and Vadali, 2014), and in this research, it is regarded as the attraction potential of a railway station – because the higher the accessibility, the higher is the attraction of passengers towards the station. Thus, the following formulation for accessibility of station i , A_i , is proposed:

$$A_i = \sum_{k=1}^K \sum_{j=1}^J \frac{\delta_{i,j,k} P_j}{I_{i,j,k}} \quad (2)$$

where,

P_j = population of the city/town that has the railway station j ,

J = total number of stations,

K = total number of trains being analyzed, and

$I_{i,j,k}$ = impedance (travel time, distance etc.) between station i and station j with train k , and

a dummy variable $\delta_{i,j,k} = \begin{cases} 1 & x < T_{i,j,k} < y \\ 0 & \text{otherwise} \end{cases}$. The time $T_{i,j,k}$ is the arrival time of train k from station i to station j .

If the train k arrives at station j anytime between desired/friendly time intervals of x and y , the dummy variable $\delta_{i,j,k}$ assumes the value of 1; otherwise, it is 0. The times x and y for a station based on the safety and convenience of passenger arrival/departure at a railway station. Convenience could include first/last mile transport to access the station and safety is ensured during particular time intervals of travel to access the station.

As an example, if x is 7 (i.e. 7 am) and y is 21 (i.e. 9 pm), a train k that arrives at station i from station j between any time after 7 am and 9 pm, the dummy variable $\delta_{i,j,k}$ is 1 and the train contributes to increasing the

accessibility of station i . For this same example, if the train k departs from station j and arrives at station i after 9 pm and before 7 am of the next day, the dummy variable $\delta_{i,j,k}$ is 0 and the train k would not contribute to the accessibility of station i . Stations that have a high number of SE train stops (arrival or departure) during the unfriendly travel periods makes those stations not so attractive for passenger travel. This could have a negative impact on the economic growth potential of the cities which the stations served. All the links among stations V that are served by train(s) form the service network. The S links of the SE train service network are divided into parts SI and SII - representing Part-I and Part-II service networks (see Fig. 1). The Part-I service network is formed when the departure time of train k at the origin station i or the train k 's arrival time at destination j is within the desired travel periods. On the other hand, in Part-II, the arrival or the departure time of the train k at the origin station i (or the destination station j) is during the odd hours of travel that may not be convenient to passengers. Thus, travel in the Part-II service network is not considered friendly from a passenger point of view. For example, under Part-II, a nighttime or early morning hour travel (as odd travel times) might not be perceived safe by a passenger due to dark or a lack of first/last mile connectivity to stations.

Therefore, based on the above narrative, for a dummy variable, say = 1 as in Part-I, it would indicate train k that serves the link connecting station i to j has an arrival or departure time at i (or at j) that is during the friendly travel periods. By default, = 0 in Part-II service network.

From the network throughput perspective, entities managing the above-discussed networks aim to utilize as many links from the infrastructure network with as many trains as possible to create more extensive service networks. Policies that allow such efforts are generally encouraged, although operational budget constraints limit those efforts.

3.2.3. Improvement or Deterioration in Connectivity

A methodology has been developed for assessing the zonal improvement (or deterioration) in connectivity (through centrality and accessibility measures). The procedure involves focusing on connectivity changes for the stations of the zone by considering the significance of train arrival/departure at the station during the friendly (or odd) travel hours compared to the 24-hour travel time scenario.

Improvement or deterioration of the SE train stations in a zone can also facilitate understanding of how the zone ranks in passenger service with respect to other zones of the Indian Railways. For example, if most of the stations in a zone show improvement in connectivity for the friendly travel time periods (compared to the 24-hour period), then the zone will have an overall improvement in the connectivity as well. Same reasoning holds for the deterioration in connectivity of a zone. Therefore, we present next the formulation to determine improvement/deterioration in centrality and accessibility for SE train stations.

We define the improvement in centrality (I_{C_i}) of a station when $I_{C_i} = (L_{i,p_{C,24-hr}} - L_{i,p_{C,d}}) > 0$

where,

$L_{i,p_{C,24-hr}}$ = centrality of station i positioned at $p_{C,24-hr}$ in ranking obtained by sorting in a descending order the centrality values of all the SE train stations in that zone for the 24-hr travel time case

$L_{i,p_{C,d}}$ = centrality of station i positioned at $p_{C,d}$ in ranking obtained by sorting in a descending order the centrality values of all the SE train stations in that zone for the friendly travel time case.

Similarly, we have, for the accessibility (I_{A_i}) of a station when $I_{A_i} = (L_{i,p_{A,24-hr}} - L_{i,p_{A,d}}) > 0$ for the '24-hr' period and friendly travel time considerations,

$L_{i,p_{A,24-hr}}$ = accessibility of station i positioned at $p_{A,24-hr}$ in ranking obtained by sorting in a descending order the accessibility values of all the SE train stations in that zone for the 24-hr travel time case, and

$L_{i,p_{A,d}}$ = accessibility of station i positioned at $p_{A,d}$ in ranking obtained by sorting in a descending order the accessibility values of all the SE train stations in that zone for the daytime travel time case.

Similarly, the deterioration in centrality (D_{C_i}) and accessibility (D_{A_i}) of a station i occurs with the respective inequalities $D_{C_i} = (L_{i,p_{C,24-hr}} - L_{i,p_{C,d}}) < 0$ and $D_{A_i} = (L_{i,p_{A,24-hr}} - L_{i,p_{A,d}}) < 0$.

An example of the improvement and deterioration calculation for connectivity (centrality) with five hypothetical station examples has been shown in Table 1. In the example shown, it is observed that the stations A and

B exhibit improvement in ranking while stations C and D show deterioration in ranking when considering comparisons of the rankings for the Travel Period - I versus the Travel Period - II. Similar calculations can be done for station showing improvement or deterioration in accessibility rankings. The zonal improvement (or deterioration) in centrality or accessibility rankings from I to II is calculated by summing up the number of stations with improvements (or deterioration), multiplying it by hundred, and dividing the product by the number of stations in that zone. Note that the sum of the percentage improvement and deterioration in centrality for zone 1 do not add to 100% because station C did not experience any change in centrality ranking.

Table 1: Illustration of improvement and deterioration calculation with five-station examples.

Station	Zone Number	Centrality Values		Rankings*		Improvement/ Deterioration in Ranking (I to II)	Zonal Changes	
		Travel Period - I	Travel Period – II	Station Ranking Travel Period - I	Station Ranking Travel Period – II		Improvement %	Deterioration %
A	1	10	50	5	1	Improvement (from 5 to 1)	66.7%	0
B	1	20	40	4	2	Improvement (from 4 to 2)		
C	1	30	30	3	3	No change (from 3 to 3)		
D	2	40	20	2	4	Deterioration (from 2 to 4)	0	100%
E	2	50	10	1	5	Deterioration (from 1 to 5)		

* Note: The higher the centrality, the better is the ranking.

3.3. Spatial Disparity with Rail

The spatial disparity can reveal gaps in SHSE connectivity and to further justify and guide future zonal-level investment policies. Zones that have benefitted or not benefitted with SE train service during friendly (or odd) hours of travel periods are identified with the disparity analysis. This is particularly important for the Indian Railways which serves number of cities (through stations) that are economically diverse and attracting varying levels of zonal investments (IBEF, 2022). These cities also lie within their respective zones that are unevenly/unequally served by the SE trains of the Indian Railways.

Zonal disparity brought about by differences in services to stations of cities could impact the regional economic health of India. A disparity analysis could reflect any modifications, expansions, or adjustments needed in operating SE train, and to expand service connectivity among the cities within zones of the Indian Railways. The analysis could further entail identifying those cities with zones that can be made attractive with improvements in connectivity of SE trains rather than expansion in rail infrastructure.

Existing measures that can be used to evaluate spatial disparity include – dispersion indices, Lorenz curve indices, and entropy indices (Liu et al., 2020). The first two noted indices namely, the dispersion indices and Lorenz curve, cannot be decomposed to capture “between” zone and “among” zone disparities as required to understand the interactions of disparities at the zonal level. However, the Theil’s T index (an entropy index) is a decomposable measure of economic inequality and can be disintegrated into population groups, income sources or other dimensions (Cowell, 2006). In literature, the index has been used as a measure for ‘between-group’ and ‘within-group’ disparities (Liu et al., 2020), and hence, in the context of examining disparities between and among zones (as groups) of the Indian Railways in this paper, the use of this index fits the purpose.

With reference to the modified definitions of variables to the original Theil T index (Theil, 1967), the following is the formula for the index as defined below:

$$T = \frac{1}{n} \sum_{i=1}^n \frac{x_i}{\mu} \ln \left(\frac{x_i}{\mu} \right) \quad (3)$$

where,

n = number of stations from all the zones

x_i = measure of the economic potential of a station i

μ = mean economic potential measure for all n stations

The between-zone and the with-in zone inequalities are defined as $T_B = \sum_{j=1}^m s_j T_j$ and $T_W = \sum_{j=1}^m s_j \ln \left(\frac{\bar{x}_j}{\mu} \right)$

, respectively, where $S_j = \frac{n_j \bar{x}_j}{n \mu}$, m being the number of zones, n_j is the number of stations in zone j , T_j is the

Theil's T index of zone j , and $\bar{x}_j$ is the average centrality measure of zone j .

3.4. Spillover Effects of Rail Investment

The benefits of investments in rail infrastructure of a specific zone are often not limited to within that zone itself. The benefits could spatially spread across the other neighboring zones due to connectivity resulting in a spatial autocorrelation among the zones for economic opportunities. A positive autocorrelation for similarities and negative autocorrelation for dissimilarities. The autocorrelation can be described through the proximity established using a weight matrix considering the contingency distance between two zones (Radzinski and Dziecielski, 2021). To identify a positive or negative spatial correlation, Moran's I could be used with a weight matrix as one of the inputs (Shi et al., 2020). The Moran's I varies between -1 (no autocorrelation) and 1 (autocorrelation). Once the spatial autocorrelation is established, an ordinary least squares regression (OLS) with Lagrangian Multiplier (LM) test can be used to determine the choice of a suitable spatial model relating the dependent and independent variables.

In spatial econometrics, the Spatial Durbin Model (SDM) has often been considered for empirical analysis to capture spatial spillovers of transportation investments and has the following formulation (LeSage and Pace, 2009; Yu et al., 2013):

$$y = \rho W y + X \beta + W X \theta + \alpha l_n + \varepsilon \quad (4)$$

where,

y is the output (e.g., job opportunities), ρ is the coefficient of spatial autocorrelation, along with α and β as parameters that need to be estimated, and ε is the error term. In this research, we have X as a vector of two independent variables – i) the investment in the local currency of the zone, and ii) the service quality of the rail transportation expressed as the zonal accessibility for the friendly and odd-hour travel times.

The SDM in Eq. 4 includes the spatially lagged y and lagged X as the independent variables to account for spillover effects. W is the spatially weighted binary contiguity matrix, which assumes that zones that share a common boundary can only impact each other's employment opportunities.

With basic transformations of terms in Eq. (4), the direct and indirect impacts can be derived by rearranging the spatial weight matrix as follows:

$$y = (I_n - \rho W)^{-1} (X \beta + W X \theta + \alpha l_n + \varepsilon) \\ \Rightarrow y = \sum_{r=1}^k S_r(W) x_r + V(W) l_n \alpha + V(W) \varepsilon \quad (5)$$

where,

$$S_r(W) = V(W) (I_n \beta_r + W \theta_r), V(W) = (I_n - \rho W)^{-1} \text{ and, therefore,}$$

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \sum_{r=1}^k \begin{bmatrix} S_r(W)_{11} & S_r(W)_{12} & \vdots & S_r(W)_{1n} \\ S_r(W)_{21} & S_r(W)_{22} & \vdots & S_r(W)_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ S_r(W)_{n1} & \cdots & \cdots & S_r(W)_{nn} \end{bmatrix} \begin{bmatrix} x_{1r} \\ x_{2r} \\ \vdots \\ x_{nr} \end{bmatrix} + V(W)l_n\alpha + V(W)\varepsilon \quad (6)$$

The average direct impacts are equal to the average of the diagonal elements of the matrix $S_r(W)$, which means that for an observation i the average impact of the dependent variables to independent variables is the coefficient of the regression represented by the average $S_r(W)_{ii}$. The average total impact is equal to the average of sum of all the rows (or columns) elements of the matrix $S_r(W)$. The average total impacts from a given row in Eq. (6) is the average total change in the dependent variable for that row for equal amount change in the independent variables across all n observations. The average indirect impacts (which represent the spillover effects) are equal to the average total impact minus the average direct impact. The Maximum Likelihood (ML) procedures can be adopted to eliminate endogeneity in estimating the parameters of the spatial model in Eq. (6) (Yu et al., 2013). LeSage's code for econometric analysis in MATLAB can be deployed for this purpose (LeSage, 2021). Thus, parameter estimates can be obtained for all three effects – direct, indirect, and total – to appropriately understand the impacts of the independent variables. A special case of SDM results when $\theta = 0$ and is called the spatial autoregression (SAR) model, while a spatially lagged X (SLX) model is obtained when $\rho = 0$ in the SDM.

4. Analysis

An analysis has been carried out to understand the inequality in connectivity and impacts of investments in the seventeen zones of the Indian Railways.

Data needed for assessing centrality and accessibility measures for connectivity were the spatial data of the SE train stations in India and the rail network of the Indian Railways. These were obtained from various online sources (including Mapcruzin.com, 2021; Indian Railways, 2021). The 17 zones of the Indian Railways were constructed on a geographical information system (GIS) software with the railway zonal map from the MapsofIndia (MapsofIndia, 2021) and corroborated with official webpage of various zones from the Indian Railways (Indian Railways, 2021d).

For the year 2020, there were 1620 railway stations identified in India that served as a stop to at least one SE train operated by the Indian Railways. The city population data of the year 2011 obtained from the Census Bureau of India were used to denote population that was assumed to have access to station (2011 Census Data, 2020). The chart in Fig. 2 shows the variation of a zone's population to the number of SE train stations. The trendline indicates somewhat correlated data indicating that each railway zone has an adequate number of stations for the population that it serves with the semi-high speed and express trains. However, how well a zone with stations is relatively situated to other zones in providing equal opportunities to the passengers for semi-high speed and express train travel can only be understood with a disparity analysis – as discussed later in this paper.

The map in Fig. 3 shows the percentage of stops of SE trains across various zones of the Indian railway network for the unfriendly travel periods of 9 pm – 7 am: during these hours travel by rail becomes unfriendly especially due to lack of an adequate first/last mile transport connectivity and safety concerns of rail passengers at dark, especially in developing nations like India (Chandrasekhar, 2020). It is assumed that higher the percentage for a zone, the more unfriendly that zone is for 9 pm – 7 am travel period.

It is observed that the three zones of Eastern Railway, South Eastern Railway, and South East Central Railway have the largest percentage of about 64-70% of SE train stops during the 9 pm – 7 am travel periods. The Western Railway, North Western Railway, and the South Central Railway have the lowest percentage of about 52-54% SE train stops during the same travel periods. The map in Fig. 3 also shows the spatial location of the top-three stations (cities) with the highest frequency of stops during the unfriendly travel period. Some of the major stations are Kanpur Central, Vijayawada Junction, Jhansi, etc., as shown in Fig. 3.

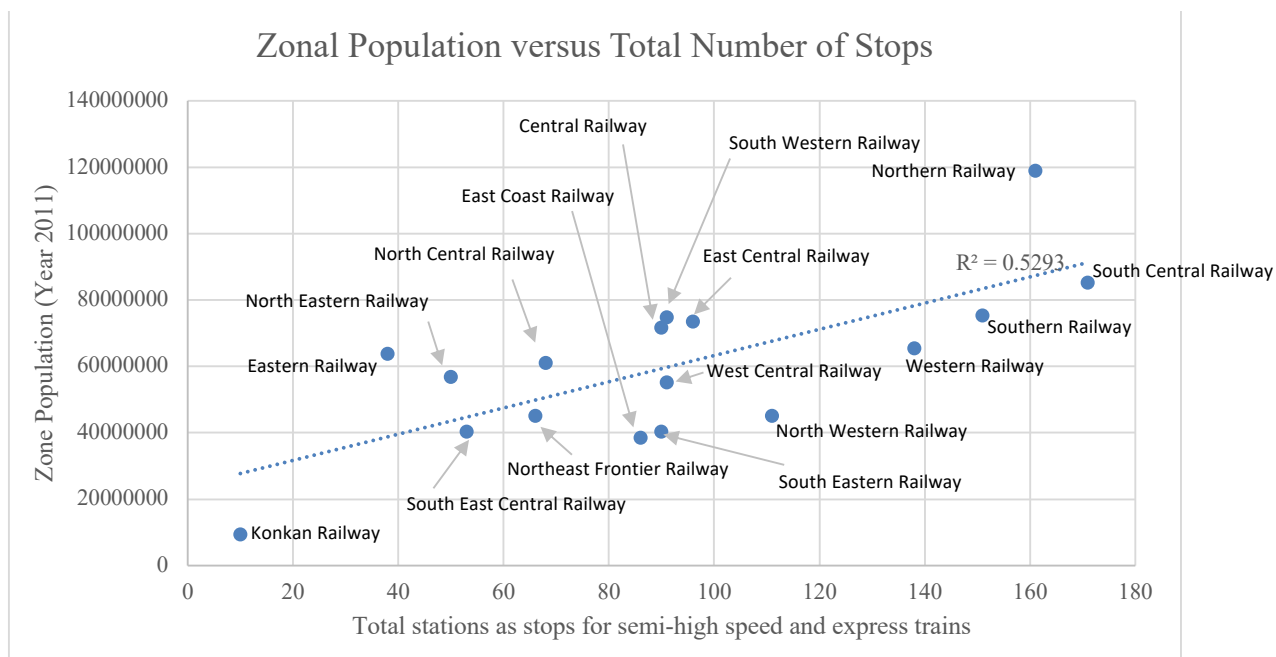


Figure 2: Variation in total number of stations versus population in the 17 zones.

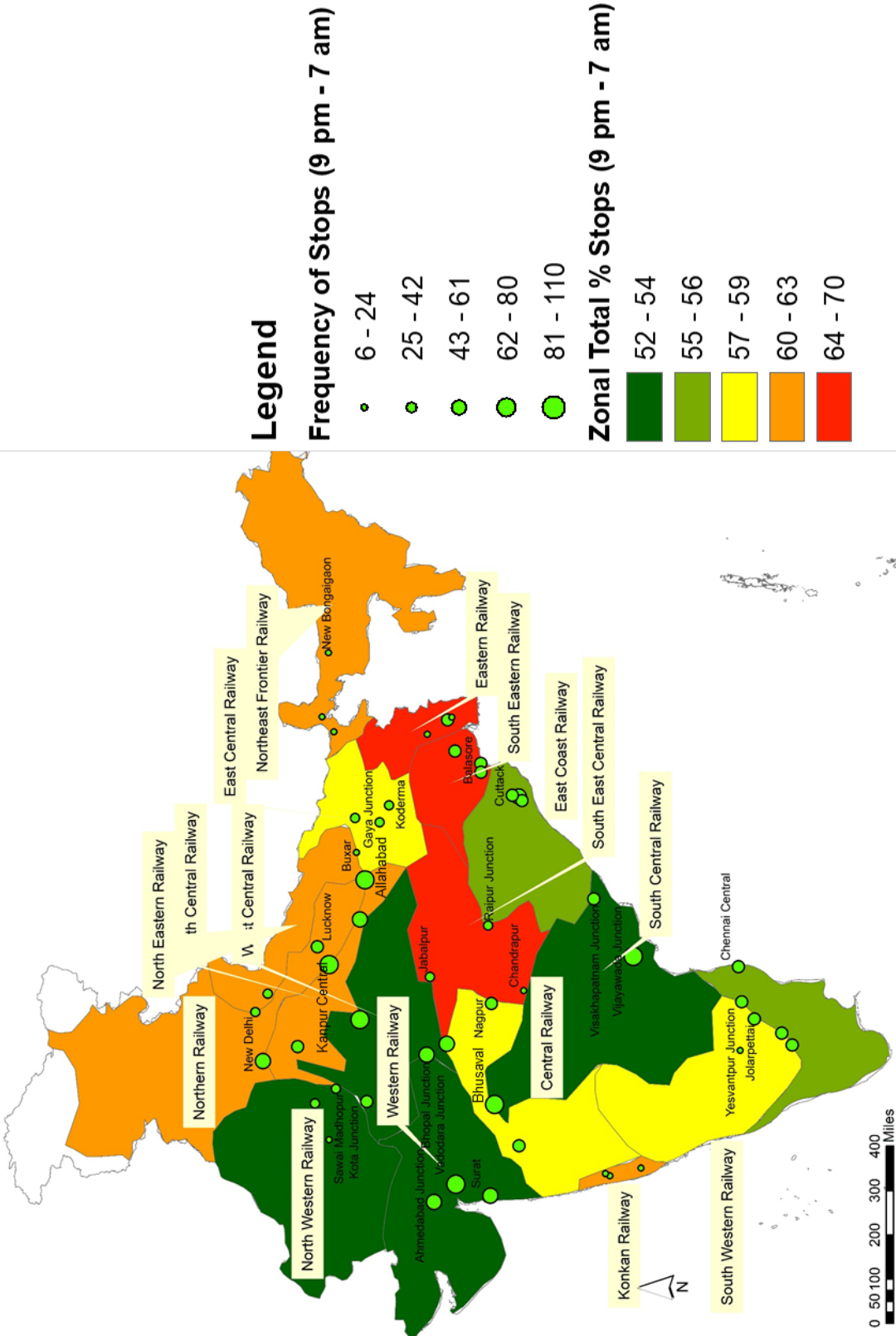


Figure 3: Frequency of train stops between 9 pm – 7 am (nighttime) across stations and zones

4.1. Centrality and Accessibility Outcomes

The percentage of stations with improvement (and deterioration) in centralities (and accessibilities) were evaluated for the seventeen zones. The evaluations were for the two connectivity measures for the 7 am – 9 pm period relative to the connectivity in the 24-hour period with the 9 pm – 7 am travel period considered to be unfriendly for passengers. Therefore, the purpose of comparing the 24-hour period with the 7 am – 9 pm period is to get a comparison between the overall connectivity for the 24-hour period and the actual connectivity that is offered to the rail passengers during the time period 7 am – 9 pm considered as friendly. This would highlight zones attracting passengers during the friendly 7 am – 9 pm travel periods. Further, the significance of identifying these zones is that the policy decisions geared to increase zone connectivity by introducing a new SE train service could consider any centrality or accessibility impacts during friendly 7 am – 9 pm travel periods of passengers.

The population data required for the calculation of accessibility of stations (and zones) for the five-year period 2016 to 2020 were obtained by assuming an average population growth of 1% per year of India from the World Bank and using the 2011 Census year population data as the base year (The World Bank, 2022).

The degree centrality and accessibility computed for each station were spatially mapped to the station's zone. From the maps in Figs. 4 and 5, it is observed that in the south, the zones of the South Central Railway, South Western Railway, and Southern Railway and, in the north, the zones of the Northern Railway and the North Central Railway - would have the highest percentage of stations with improvement in centrality for the 7 am – 9 pm travel period relative to 24-hour period centrality. This percentage improvement would be in the range of 44 to 50%. The lowest percentage of stations with improvement in centrality would be in the Eastern Railway and the Konkan Railway zones of around 20-24%.

The Konkan Railway zone stations of about 20-27% will be the lowest centrality deterioration for the 7 am – 9 pm period compared to the 24-hour period. On the other hand, several zones with a high percentage of stations would deteriorate in centrality for the 7 am – 9 pm period compared to the 24-hour period. These zones include the Western Railway and North Eastern Railway, and the others, with about 49-55% of stations - as shown in the map of Fig. 4 with the red color shade. This means that connectivity seems to reduce in these zones, with the high percentage of stations noticing a drop in centrality during the friendly 7 am – 9 pm travel periods.

Comparing the two maps in Fig. 4, it is observed that the two zones of the Northern Railway and the Southern Railway have the most significant percentage of stations that would experience changes (improvement and deterioration simultaneously) in centrality for the friendly 7 am – 9 pm travel periods compared to the 24-hour period. This observation could be due to a significantly high number of SE trains that terminate or originate at the New Delhi railway station in the Northern Railway zone and the Chennai Central railway station in the Southern Railway zone.

For the accessibility case, the North Western Railway, Western Railway, and South East Central Railway zones would have the highest percentage of stations experiencing improvement for the friendly 7 am – 9 pm period (see Fig. 5). These percentages would be in the range of 50-52%. The zones of the East Central Railway, Eastern Railway, Konkan Railway, and Southern Railway would have the lowest percentage of stations with improvement in accessibility for the same travel periods. The largest percentage of stations experiencing deterioration in accessibility for this travel period would be for the South Western Railway, Southern Railway, East Coast Railway, South Eastern Railway, and Northern Railway. The percentage would be around 53-55%.

The Eastern Railway and the Konkan Railway would have the lowest percentage of stations (around 40-43%) that would experience deterioration in accessibility. The Southern Railway zone, among all the other zones, would have the lowest percentage of stations with accessibilities that would deteriorate and improve in the 7 am – 9 pm period compared to the entire 24-hour period. The total percentage of stations that would exhibit changes (improvement and deterioration) in connectivity for a zone does not add to 100%. Some of the stations in any zone would not change in centrality or accessibility for the 7 am – 9 pm travel periods.

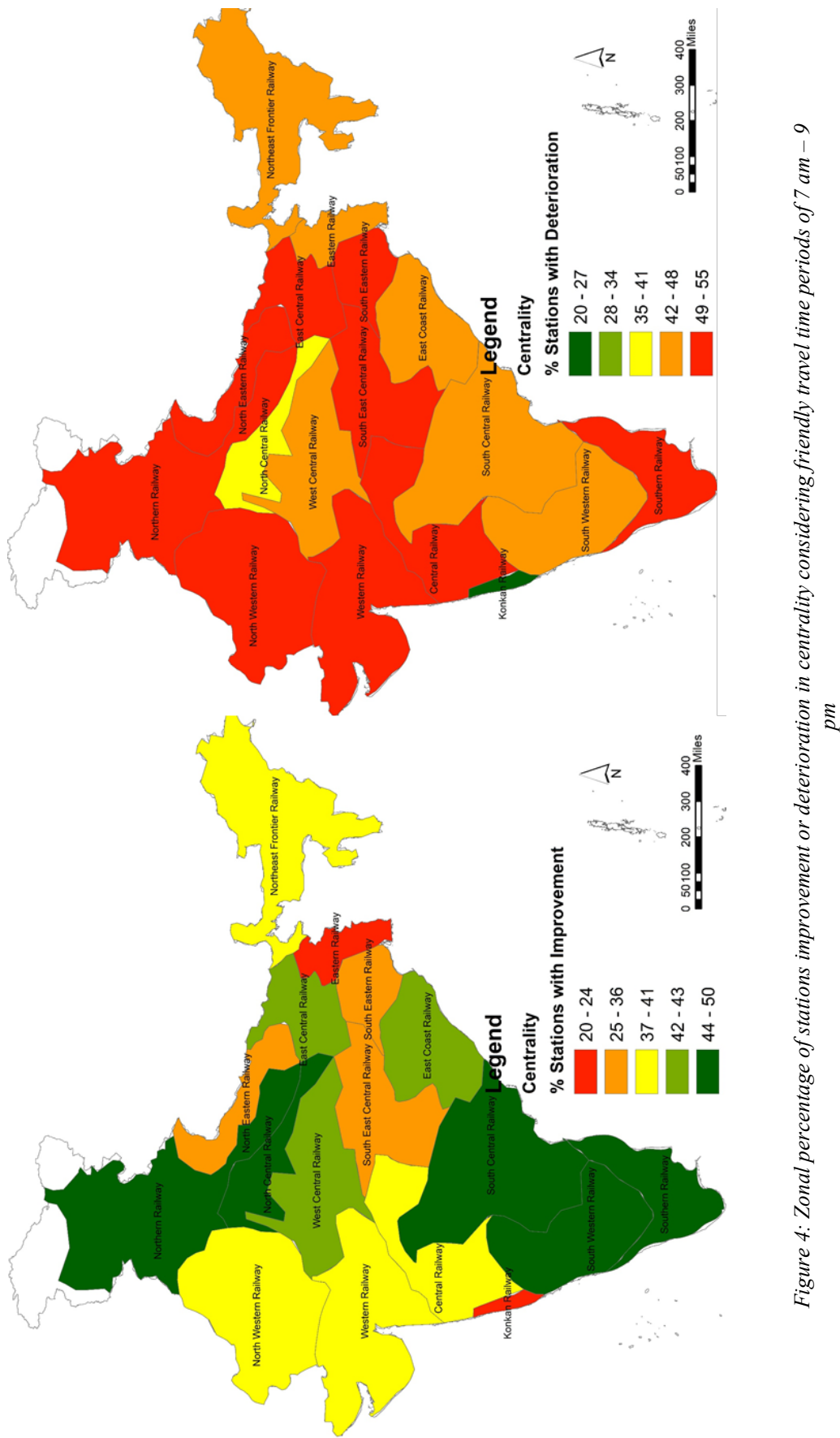


Figure 4: Zonal percentage of stations improvement or deterioration in centrality considering friendly travel time periods of 7 am – 9 pm

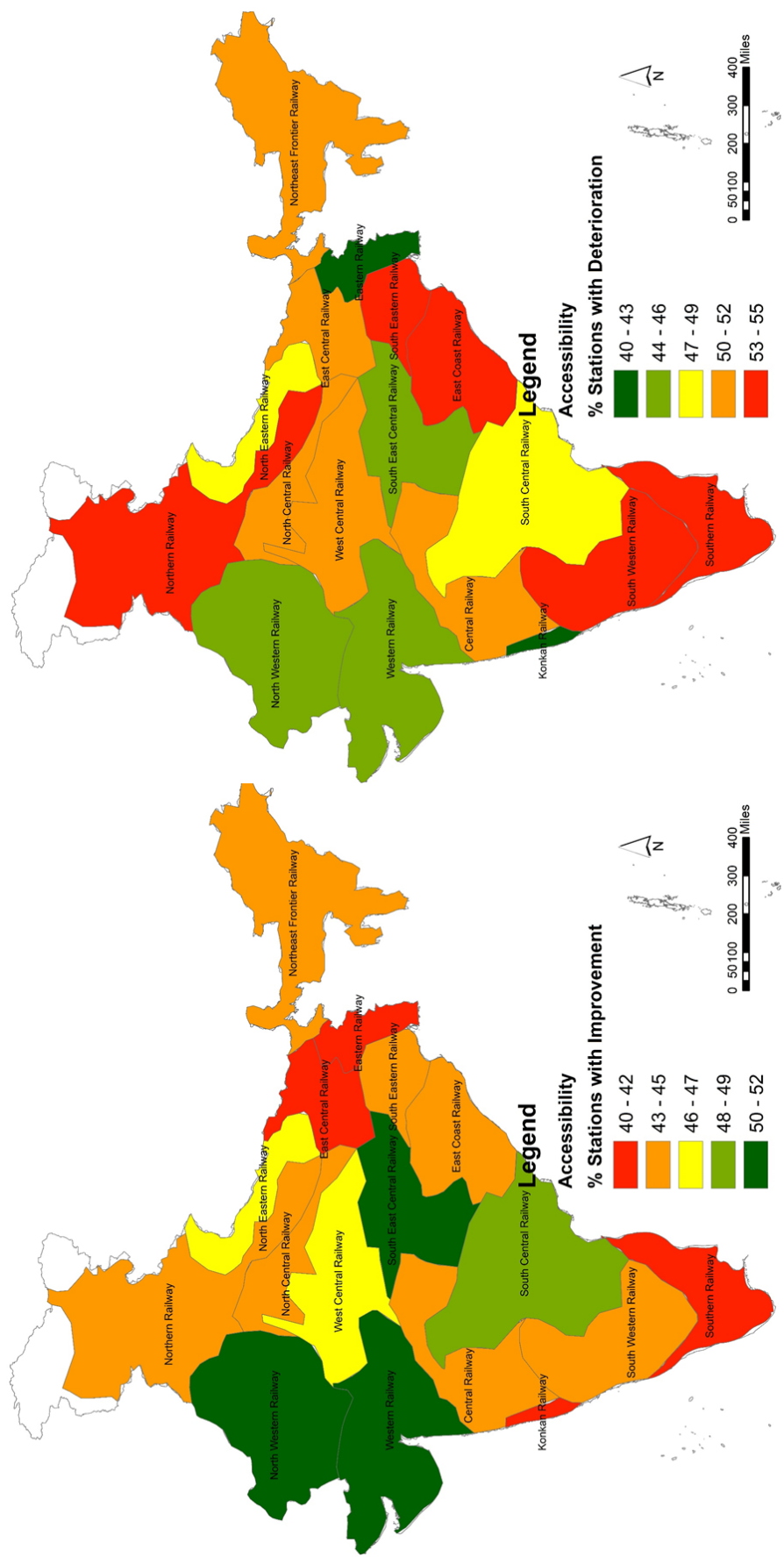


Figure 5: Zonal percentage of stations improvement or deterioration in accessibility considering friendly travel time periods of 7 am – 9 pm

The mean centrality and accessibility of stations in a zone were normalized as percentage with respect to the highest zonal centrality and accessibility and presented in the maps of Figs. 6 and 7, respectively.

It was observed that the Central Railway exhibits the largest normalized percentage centrality compared to the other sixteen zones for both the 24-hour period and 7 am – 9 pm period (see map in Fig. 6) - which could be because several trains arrive or depart at the stations of this zone than at the stations in the other zones and as corroborated with the Indian Railways timetable for the SE trains and an independent website on running status of trains (Running Status, 2021; Indian Railways^{2021c}). Thus, on average, a station in the Central Railway zone is more connected than the stations in other zones. The Northeast Frontier Railway and the North Western Railway zones have the lowest normalized centrality for the 24-hour period and the 7 am – 9 pm period, respectively. Thus, it appears that the stations in these two zones have the lowest SE train connectivity compared to the other zones in those two respective time periods. It is also noted that there was an almost equal distribution of normalized centralities among majority of the zones for the two time periods. This could indicate that stations in most zones would do not experience changes in centrality (connectivity) for the two time periods.

Based on the findings as illustrated using the map in Fig. 7, the zone of North Central Railway has the most significant normalized accessibility for both the 24 hour period and the 7 am – 9 pm period. The high accessibility value for North Central Railway zone is attributed to an increased number of SE trains for the two analyzed time periods. This is also supplemented by the proximity of the stations in these two zones to cities in which they lie with a high population, thus eventually contributing to large values of average accessibility.

Therefore, stations in the North Central Railway zone have an increased number of SE train stops than the other stations in other zones both during the 24-hour period and the 7 am - 9 pm period.

It is observed that several zones including the South Western Railway, East Coast Railway, Western Railway, and the North Frontier Railway have decreased in SE train accessibility from the 24- hour period to the 7 am – 9 pm period. This could be expected because the number of train stops are reduced during the 7 am – 9 pm period compared to the total 24-hour period. But this also indicates that a significant number of SE train stops are during the unfriendly 9 pm – 7 am period.

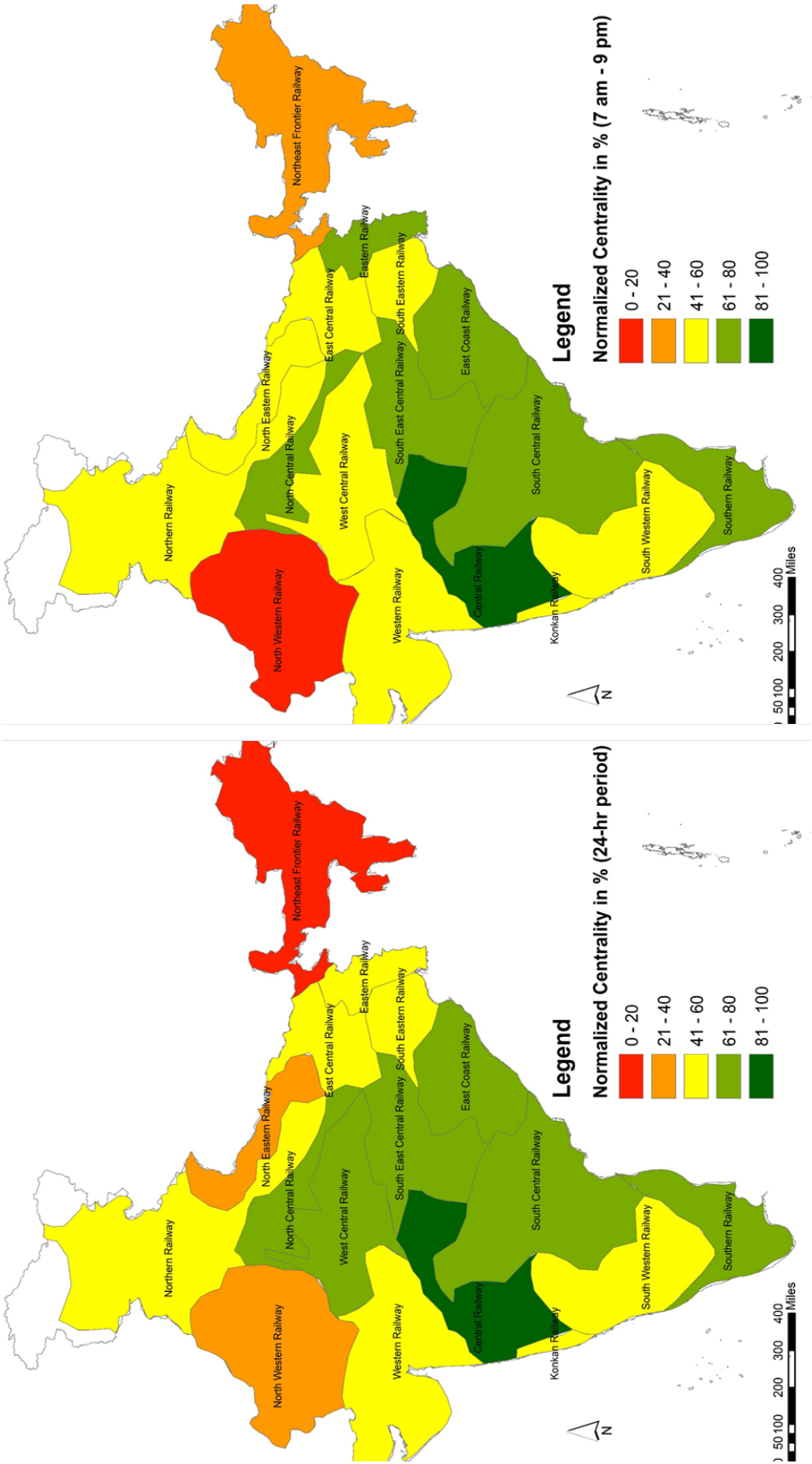


Figure 6: Normalized centrality in percentage for zones for the 24-hr and the 7 am – 9 pm periods

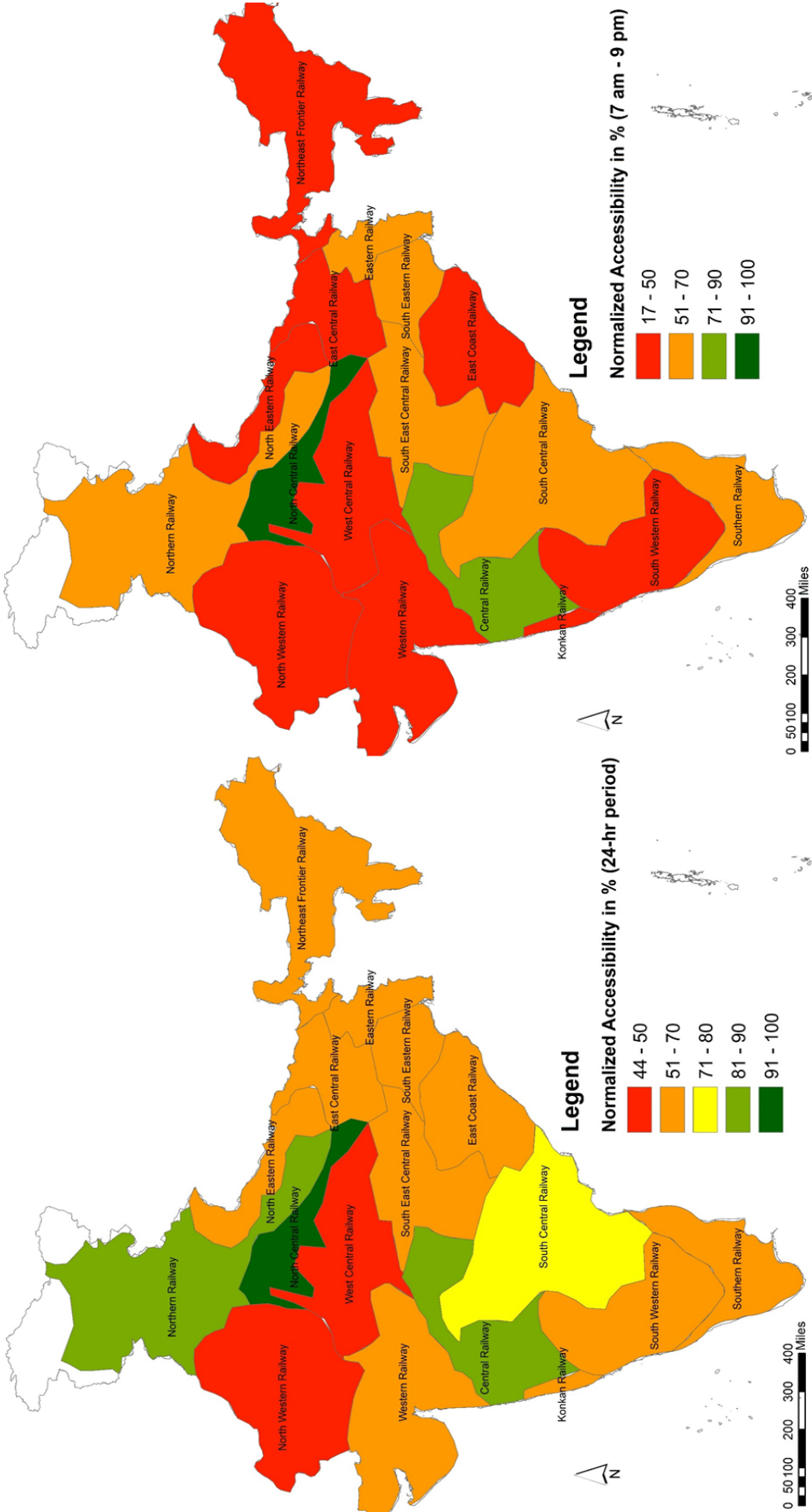


Figure 7: Normalized accessibility in percentage for zones for the 24-hr and the 7 am – 9 pm periods

4.2. Disparity Analysis

The disparity analysis showed that if only the degree centrality is considered in evaluating inequality in connectivity among zones, the South Eastern Railway and the Eastern Railway have the largest Theil's T index (equal to 1 each) for the 24-hour period (see Fig. 8). Thus, these two zones are the least connected with SE trains for the 24-hour period considered. This also suggests that if the 24-hour period of SE trains is considered, the stations in these two zones could offer a reduced economic opportunity to the population they serve, compared to those in other zones.

For the friendly 7 am – 9 pm travel periods, the South Central Railway zone has the largest inequality in centrality (Theil's T index equal to 0.9) as compared to other zones of the Indian Railways. Thus, indicating that the scheduled stops of SE trains between 7 am to 9 pm at the stations of this zone is not much of a utility with reduced connectivity and minimal opportunities for economic growth in this time period. In comparison to the other zones, the Konkan Railway with the lowest indices for centrality for the 24-hour period (equal to 0.41) and the 7 am – 9 pm travel period (equal to 0.4) has the highest potential for economic growth with SE train service that its stations receive.

Based on a similar analysis with accessibility, the zones of the South East Central Railway and the Southern Railway exhibit the largest inequality for the 24-hour travel period with T index equal to 0.62 and 0.59, respectively. This shows that the population in these two zones has an overall reduced opportunity for economic growth than other zones for the 24-hour period with the SE train connectivity. For the 7 am – 9 pm travel period, with Theil's T index of 1.75, the South East Central Railway zone (compared to the other zones of the Indian Railways) also has the least opportunity for economic growth with the SE train connectivity.

The zone of Konkan Railway, compared to other zones, has the smallest Theil's T index equal to 0.09 and 0.58 for the 24-hour and the 7 am – 9 pm travel periods, respectively, and thus, has the largest potential for economic growth with SE train connectivity.

With the decomposition analysis for the 'between-zone' inequality and 'within' zone inequality for the two periods presented in the chart of Fig. 9, it is evident that there is a large disparity in the centrality of stations between different zones (with Theil's T index equal to 0.82 for the 24-hour period and 0.71 for the 7 am – 9 pm travel period) than within the zones (with Theil's T index equal to 0.01 for both the periods). Therefore, for the two time periods the connectivity provided by SE trains is highly unequal between zones. Therefore, while this poses an economic advantage for some zones, other zones do not have the same advantage because of unequal opportunity.

When analyzed for disparity using accessibility, the between-zone inequality is higher for the service times 7 am – 9 pm travel period (with Theil's T index equal to 1.03) as compared to the 24-hour travel period (with Theil's T index equal to 0.36). However, the index for these periods is higher than the Theil's T index of 0.03 for the 24-hour period, and the index of 0.09 for the 7 am – 9 pm travel period for the 'within' zone inequality of accessibility. This indicates that the economic opportunities from travel with SE trains considering accessibility (as a surrogate of connectivity) are unequal for the population with-in the zones compared to between zones for the two time periods.

Overall Disparity (Theil's T Index)

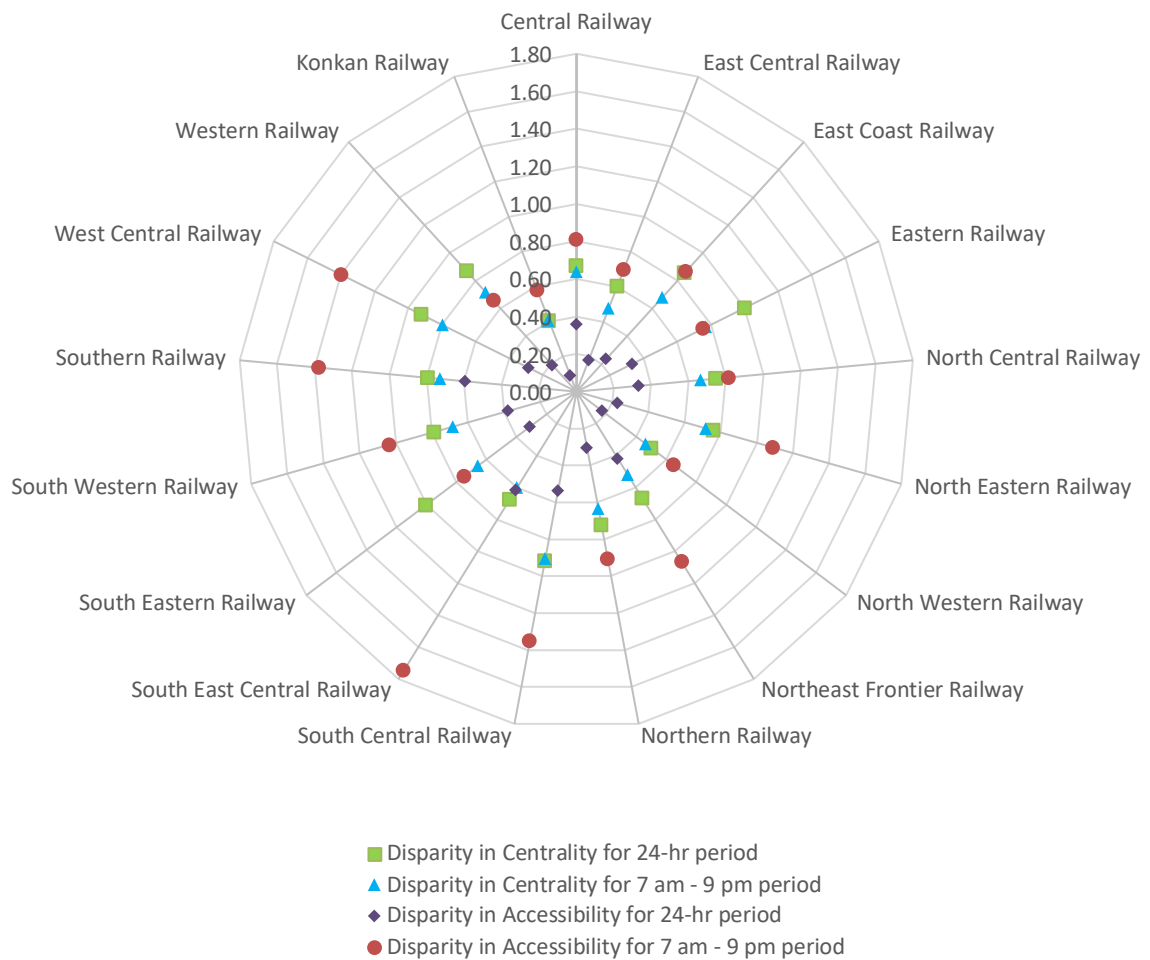


Figure 8: Overall disparity outcomes for with centrality and accessibility measures



Figure 9: Between-zone and with-in zone disparity analysis

5. Investment Policy and Other Connectivity Implications

The Indian Railways provide direct funding for the operation and maintenance of rail infrastructure and services. One of the outcomes that need investigation relates to how these investments have been effective in the zones' economic growth, which would alleviate disparities that currently exist between them. Therefore, we discuss the impact of investment by the Indian Railways on employment in zones in the past five years. This discussion is necessary to see if the efforts in investment by the Indian Railways have facilitated increased employment in zones. Further, we also discuss how station accessibility with SE train arrival/departure delays in a zone during friendly and unfriendly travel periods are also causal to employment impacts in the past five years.

5.1. Investment in Indian Railways

The data on investment in passenger rail by the Indian Railways were gathered from the Indian Railways webpage (Indian Railways, 2021c). The chart in Fig. 10 shows that the investment has been varying over the last five years

(from 2016 to 2020) across the zones but consistently increasing for a given zone. Data from Indian Railway budget show that the level of investment in each zone has been proportional to passenger rail revenue generated in a zone over the same period - although a major percentage of the revenue generated for Indian Railways is from goods transportation. These investments are primarily accounted for as total working expenses and include repairs and maintenance of permanent way and works, carriages and wagons, operating expenses (rolling stock and equipment, traffic, and fuel), staff welfare and amenities, etc. (Indian Railways, 2021a). While some of the zones in recent years have a higher than 100% operating ratio, the ratio is lower than 100% for zones such as East Coast Railway and North Central Railway – which indicates that not all zones are profitable for the Indian Railways. However, very little is known if investment in rail changes economic growth or impacts job opportunities in any way within the zones.

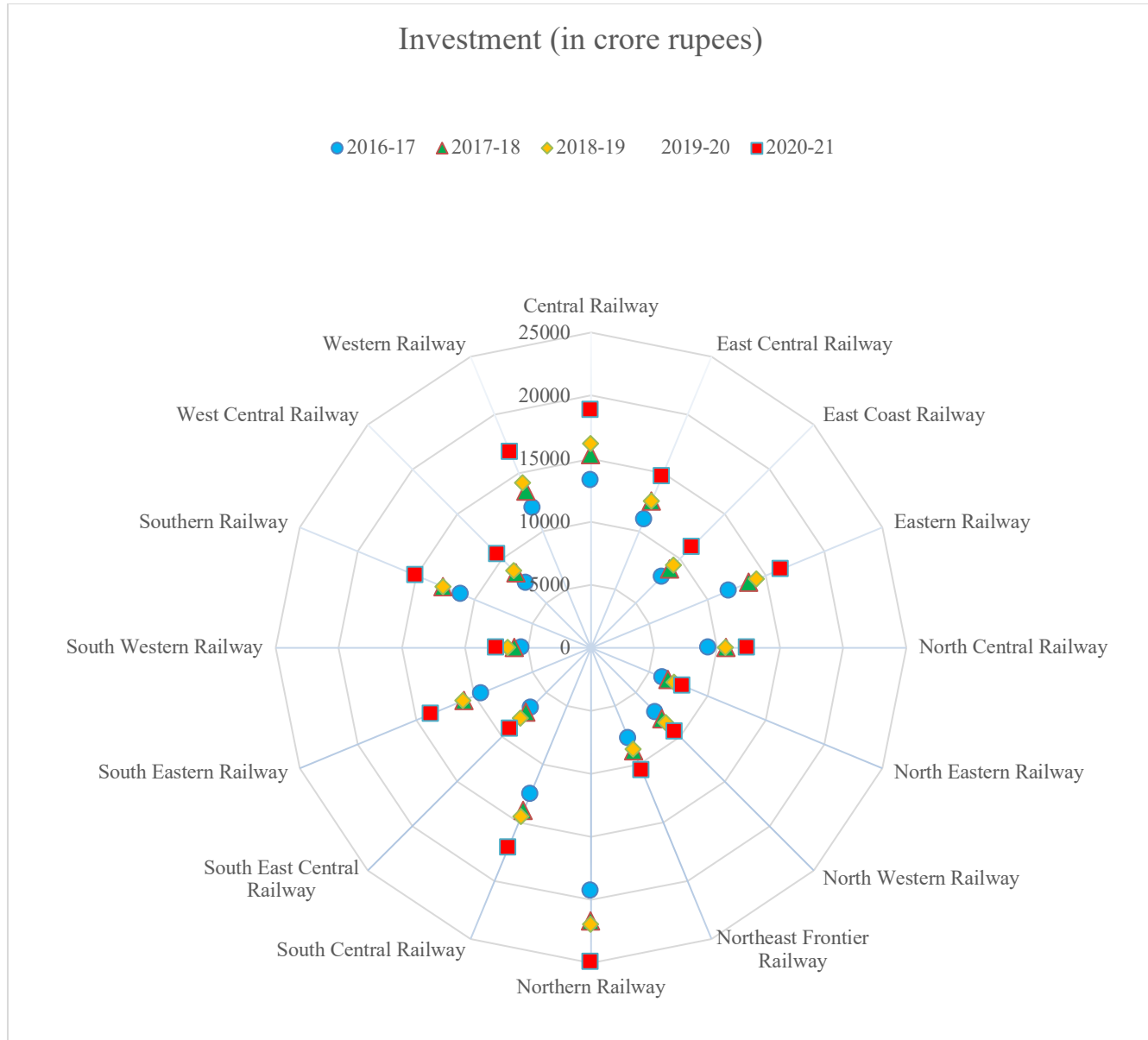


Figure 10: Annual zonal investments by the Indian Railways (Source: Chandra and Narayanaswami, 2024)

5.2. Train delays

Accessibility (compared to centrality) can be considered a suitable indicator of SE train connectivity for the Indian Railway zones. This is because accessibility is sensitive to the city population served by the SE train stations and impedance (travel time). Therefore, further insights were drawn on the impact of zonal accessibility (through SE trains) on the zones' economic growth and the investment undertaken by the Indian Railways.

We specifically focus on accessibility impacts for the two time periods of analysis: 7 am – 9 pm, and 9 pm – 7 am. While the 7 am – 9 pm travel period is considered friendly to passengers, 9 pm – 7 am is deemed unfriendly due to a lack of first/last mile connectivity or safety reasons.

For these two analysis periods, yearly SE trains running status gathered from Cleartrip.com from 2016 to 2020 showed the trains' impunctuality (delays) at various stations of the zones. The variation in the annual delays of these trains at the zonal level is shown in Figs. 11 and 12.

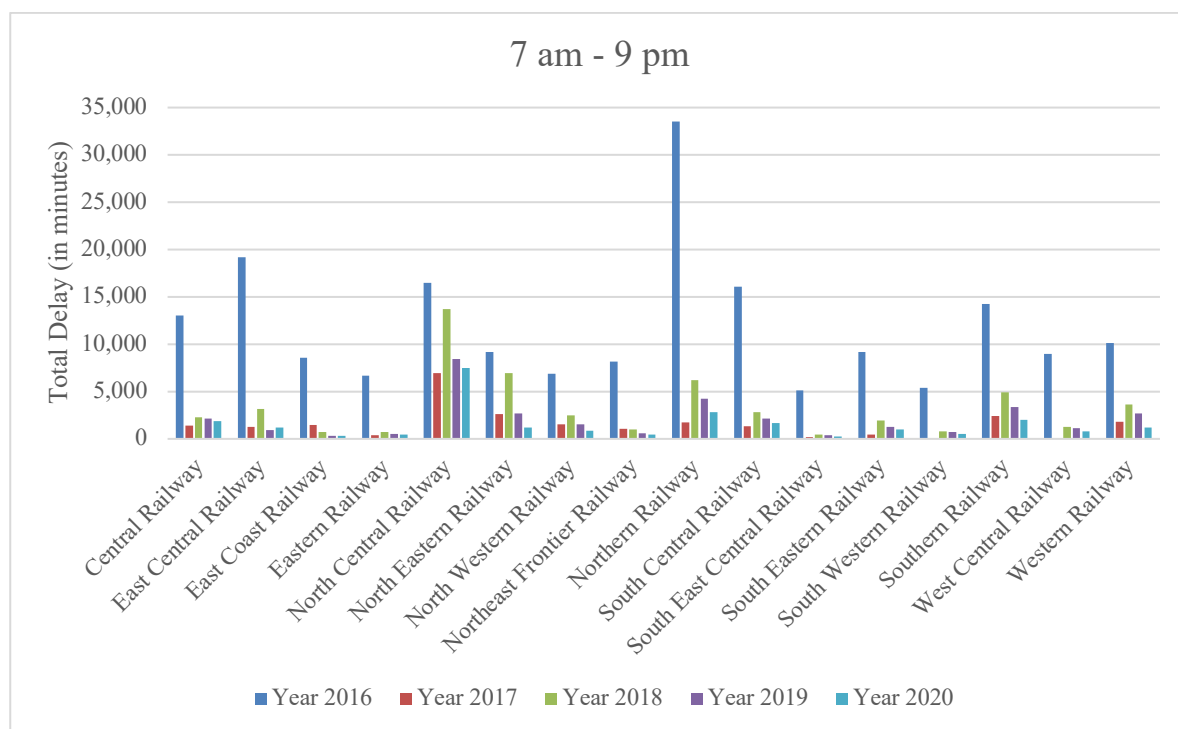


Figure 11: Annual delay variation for SE trains across zones for train stops for the 7 am – 9 pm period

The total annual SE train delay for 2016 was the highest for the Northern Railway zone for the 7 am – 9 pm period, exceeding 30,000 minutes. Other zones in that year had a higher delay than in the other years. The zones of the East Central Railway, East Coast Railway, North Eastern Railway, North Western Railway, Southern Railway, and Western Railway had the lowest delays in 2020. The rest of the zones experienced the lowest delays in 2017 for the 7 am – 9 pm period.

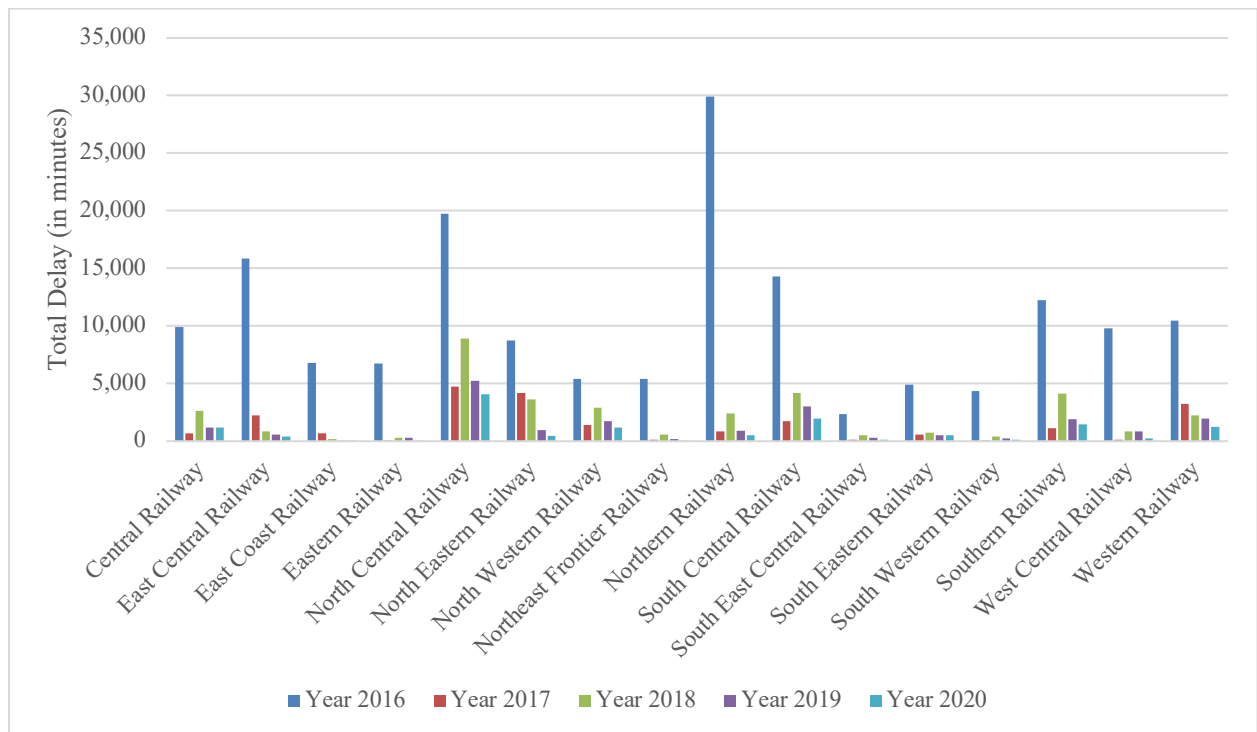


Figure 12: Annual delay variation for SE trains across zones for train stops from 9 pm – 7 am

For the 9 pm – 7 am period, the total annual delays were the highest in 2016 for all the zones, with the Northern Railway zone experiencing the largest delay (about 30,000 minutes) compared to the other zones. The zones with the lowest delays in 2020 were the East Central Railway, East Coast Railway, North Eastern Railway, North Western Railway, South Eastern Railway, South Western Railway, and Western Railway. The rest of the zones had the least delays in 2017 from 9 pm – 7 am.

Accessibility for each station was calculated by considering the impedance as the sum of the actual origin-destination travel time of SE trains and the same origin-destination pair delay. Thus, variations in the zonal accessibilities using stations' city population were calculated for 2016-2020. The population for each of the years from 2016 to 2020 was calculated using the base census year population of 2011 and extrapolating with a population growth rate of 1% obtained from the World Bank (as noted in the previous section). A zone's accessibility equal to the mean station accessibility in a zone was calculated for each of the five years for 2016-2020 for the two periods: 7 am – 9 pm, and 9 pm – 7 am. Thus, 80 data points for the five years (2016-2020) were obtained on accessibility for each period for the sixteen zones (excluding the zone of Konkan Railway, which did not have information on investment by the Indian Railways).

5.3. Spillover effects

Spillover effects of investment and SE train service connectivity on employment across zones were analyzed. Data on employment (job opportunities created) from 2016 to 2020 for zones were used as the dependent variable, while the investment and the zonal mean accessibility were used as the independent variables. Each Indian Railways zone is made up of several districts, that have information on total workers (indicative of the total employment) as gathered from the Office of the Registrar General & Census Commissioner, India for the year 2011 (Census – 2011, 2021) and these were aggregated to yield employment data. Further extrapolation was carried out using employment growth rate from 2012-2019 (Trading Economics, 2021) and the five-year employment data from 2016 to 2020 were prepared.

The distances among centroid of zones varied significantly. Our focus was on investigating spatial impacts on job opportunities among zones that were immediate neighbors to each other. We chose to consider a queen-type

measure of accounting two zones as neighbors if there was any common share of their boundary required to calculate spatial autocorrelation. GeoDA, a popular open-source spatial analysis software, was used to calculate the Moran's I and conduct the LM tests (GeoDA, 2021). In our study, the value of Moran's I was positive and significant at 1% for the employment distribution (as job opportunities) of zones. LM test output indicated that a spatial lag model could be used for further analysis for the job opportunities (used as the dependent variable) and the zonal investment and accessibility with odd hours of travel used as the two independent variables. The LM test suggested the suitability of a spatial autoregressive (SAR) model for our case with total zone employment (y) as the dependent variable.

Findings from analysis show that the impact of investment on job opportunities in zones are significant at 1% level with estimates of 43.206. Limiting the findings just to the coefficients estimated for investments by the Indian Railways does not solely reflect their marginal effects on job opportunities; therefore, assessing its average direct and indirect effects become necessary. The estimations are based on the concept of spillover effects introduced in the earlier section.

The investment had positive direct and indirect effects. The estimates are found to be 54.392 (at 1% significance level) and 115.318 (at 5% significance level) for the direct and indirect effects (spillover effects), respectively. The total effect was 169.711 (at 5% significance level). The above findings are summarized in Table 2. Thus, the policy implications of the findings are that the Indian Railways could continue to invest in sustaining job opportunities in the 17 zones.

The mean accessibility of zones during the 7 am – 9 pm period positively impacted the job opportunities with weight equal to 0.020, the direct impact weight being 0.025, the indirect impact weight being 0.054 – all at 10% significance level. However, the weight of the total impact of mean accessibility on job opportunities was not significant. Therefore, based on the analysis, there is some evidence that a zone's job opportunities by SE train connectivity during the friendly 7 am – 9 pm travel periods are impacted by the neighboring zones' accessibilities with the direct and indirect effects.

The mean accessibility of zones during the 9 pm – 7 am period is significant at a 5% significance level with a (negative) weight of -0.033. This shows that with the increase in the 9 pm – 7 am accessibility, there have been a reduction in the zonal employment. The direct impact weight of the accessibility in this period being a -0.043 at 5% significance level indicates that this accessibility has contributed to a decrease in job opportunities in a zone. Further, the indirect effects (spillover effects) of 9 pm – 7 am accessibility also lower job opportunities in neighboring zones. The indirect impact is -0.091 at 5% significance level, and the total impact weight is -0.133 at 10% significance level. These findings indicate that the accessibility during this period have been detrimental to the job opportunities in the zones. Thus, policies that would increase the 9 pm – 7 am accessibility provided by the Indian Railways could increase job opportunities in the zones.

The spatial autocorrelation coefficient (ρ) is found to be 0.723 and spatial dependence for job opportunities is observed at 1% significance level. This indicates that a zone is affected by job opportunities in neighboring zones.

Table 2: Estimation results of SAR for the Indian Railway zones

Variable	Weight	Direct effect	Indirect effect	Total effect
Investment	43.206*** (11.762)	54.392*** (8.448)	115.318** (2.289)	169.711*** (3.043)
Mean Accessibility (7 am – 9 pm)	0.020* (1.357)	0.025* (1.344)	0.054* (1.099)	0.079 (1.195)
Mean Accessibility (9 pm – 7 am)	-0.033** (-2.310)	-0.043** (-2.261)	-0.091** (-1.573)	-0.133* (-1.806)
ρ	0.723*** (9.449)			
R^2	0.9911			

Note: The numbers in parenthesis denote t-statistics

* Significance level 10%

** Significance level 5%

*** Significance level 1%

Recent thrust of the Indian Railways has been on soliciting private participation in rail connectivity and capacity augmentation projects (Indian Railways, 2021b). The underlying policy goals of these efforts include supplementing government investment in rail projects by private capital flows and matching the transport capacity needs with the expected GDP growth of the nation. Indian Railways anticipates maximizing land use and spurring economic activities around stations in future with assistance from private entities under a public-private partnership (PPP) model (BusinessLine, 2021). However, complete privatization of the Indian Railways has been forewarned and call for a check on productive efficiency, particularly ensuring an equity-allocation (Kuriakose and Gupta, 2021).

PPPs in projects of the Indian Railways have been evolving over last several years in India, including operating partnerships with state governments and some PPPs in the past have been discontinued as well (Raghuram and Gangwar, 2010). All these efforts indicate Indian Railways is keeping a foreword approach to revitalizing poorly performing elements of the passenger rail operations in India.

To enhance connectivity across the zones, the Indian Railways had introduced over 800 new passenger trains in the last five years, out of which 67 new express trains were rolled into service in 2014-2015 (Economic Times, 2022a). Currently, there are two fully operational semi-high speed Vande Bharat trains that terminate and originate at New Delhi railway station, which is in the national capital of India and lies in the most populated zone of the Northern Railway (IRCTC, 2022; 2011 Census Data, 2020). As per the Union Budget, there are also plans to introduce 400 new Vande Bharat semi high-speed trains between other major stations/cities of India in the next three years (Economic Times, 2022b).

6. Concluding Remarks

This research investigated the impact of SE train connectivity on seventeen zones of the Indian Railways. Degree centrality and accessibility were used as the two measures of connectivity. Further, evaluations were carried out to assess the inequality among zones and understand how the past five years of accessibility variations and investment policy of the Indian Railways have played a critical role in economic growth (in terms of job opportunities) of the zones.

In this paper, the analyses focused on two time periods: a 24-hour period and the 7 am – 9 pm period. The comparative analysis between the two periods highlighted the SE train connectivity during the 7 am – 9 pm period relative to the daily connectivity of the 24-hour period. The 9 pm – 7 am travel period is considered unfriendly for rail passenger travel in India due to the lack of adequate first/last mile connectivity and safety concerns in accessing rail stations at night or early wee hours of travel. Therefore, the effective connectivity from the SE trains would be during the 7 am – 9 pm travel period.

An analysis carried out with the early year 2020 SE train connectivity of the Indian Railways showed a more substantial disparity in their connectivity among its seventeen zones than within a zone for the two periods. The Theil's T index for the between-zone disparity was much higher than the within-zone disparity for both centrality and accessibility measures – indicating that the inequalities in connectivity exist between zones more than between the stations (as cities) in a zone.

The South Central Railway zone has the largest inequality in centrality as compared to other zones of the Indian Railways for the 7 am – 9 pm period, and the two zones of the South East Central Railway and the Southern Railway exhibit the same for the 24-hour travel period. The Konkan Railway ranks low in disparity for both 7 am – 9 pm period and the 24-hour period. For accessibility, the South East Central Railway zone exhibits the largest inequality for both travel periods. For both periods, it was found that the Central Railway has the largest mean centrality, and the North Central Railway zone has the largest mean accessibility of all the stations in the respective zones.

Positive spillover effects of SE train accessibility on job opportunities in zones are observed during the 7 am – 9 pm period, which suggests that policies that increase the zonal connectivity through new SE trains could be promoted by the Indian Railways. This would mean that a zone lagging behind other zones in economic growth could have an increased SE train connectivity – particularly to those with high employment. At the same time, although SE train accessibility could increase during the 9 pm – 7 am period, due to a negative spillover effect, SE train connectivity may not add to the increase in employment at the zonal level. In this research, it was also observed the positive impact of Indian Railways investment in various zones and the subsequent increase in employment. This indicates that for

sustained economic growth in zones, the Indian Railways should continue to invest in the rail network infrastructure (including tracks, rolling stocks, etc.).

In summary, two important suggestions are made based on the research findings on SE train connectivity - first, the Indian Railways should focus on identifying zones for investments that have the most considerable disparity in centrality and accessibility as these two are indicators of the economic potential of a zone, and second, to enhance the service quality offered by the SE trains, their halts at critical stations for passenger boardings/alighting during nighttime and early morning hours should be facilitated with increased first/last mile connectivity and enhanced safety through streetlights leading to the stations. In the absence of these steps, the economically prosperous zones will continue to advance in employment growth, resulting in increased disparity with marginalized/less strategically served zones of the Indian Railways.

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